EVERY SEPARATED ETALE V' -> V OVER U YIELDS A PUSHOUT SQUARE THAT IS AN ELEMENTARY DISTINGUISHED SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1763-1778 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 1780-1811, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 22 of queue_spaces-pushouts.json (pb=1369, stmt 1763-1778, proof 1780-1811).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04S6: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1811–1811. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DVI: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1724–1735. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let V, U be algebraic spaces over S. Let $V' \subset V$ be an open subspace and let $f' : V' \to U$ be a separated étale morphism of algebraic spaces over S. Then there exists a pushout*

$$\begin{array}{ccc} V' & \longrightarrow & V \\ \downarrow & & \downarrow f \\ U & \longrightarrow & W \end{array}$$

in the category of algebraic spaces over S and moreover $(U \subset W, f : V \to W)$ is an elementary distinguished square.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We are going to construct W as the quotient of an étale equivalence relation R on $U \amalg V$. Such a quotient is an algebraic space for example by Bootstrap, Theorem 04S6. Moreover, the proof of Lemma 0DVI tells us to take

$$R = U \amalg V' \amalg V' \amalg V \amalg (V' \times_U V' \setminus \Delta_{V'/U}(V'))$$

Since we assumed $V' \to U$ is separated, the image of $\Delta_{V'/U}$ is closed and hence the complement is an open subspace. The morphism $j : R \to (U \amalg V) \times_S (U \amalg V)$ is given by

$$u,\ v',\ v',\ v,\ (v'_1, v'_2) \mapsto (u, u),\ (f'(v'), v'),\ (v', f'(v')),\ (v, v),\ (v'_1, v'_2)$$

with obvious notation. It is immediately verified that this is a monomorphism, an equivalence relation, and that the induced morphisms $s, t : R \to U \amalg V$ are étale. Let $W = (U \amalg V)/R$ be the quotient algebraic space. We obtain a commutative diagram as in the statement of the lemma. To finish the proof it suffices to show that this diagram is an elementary distinguished square, since then Lemma 0DVI implies that it is a pushout. Thus we have to show that $U \to W$ is open and that f is étale and is an isomorphism over $W \setminus U$. This follows from the choice of R; we omit the details. □